The effect of vegetarian diets on plasma lipid and platelet levels.

Vegetarians have an apparent diminished risk for the development of ischemic coronary heart disease. This may be secondary to dietary effects of plasma lipids and lipoproteins, but platelets, which may also play a role, have also been observed to have aberrant functions in vegetarians. We measured plasma lipid and lipoprotein levels, platelet function, platelet fatty acid levels, and platelet active prostaglandins in ten strict vegetarians (vegans), 15 lactovegetarians, and 25 age- and sex-matched omnivorous controls. The most striking observations were a highly significant rise in platelet linoleic acid concentration and a decline in platelet arachidonic acid concentration in both vegetarian subgroups as compared with omnivorous controls. Serum thromboxane and prostacyclin levels as well as results of platelet aggregation studies did not differ among the groups tested. Cholesterol levels were significantly lower in both vegetarian groups as compared with controls, but plasma high- and low-density lipoprotein levels were lower only in the vegan subgroup as compared with omnivores. If diet produces these changes in platelet fatty acid and plasma lipid levels it may contribute to the decreased risk of coronary heart disease and possibly atherosclerosis in vegetarians.